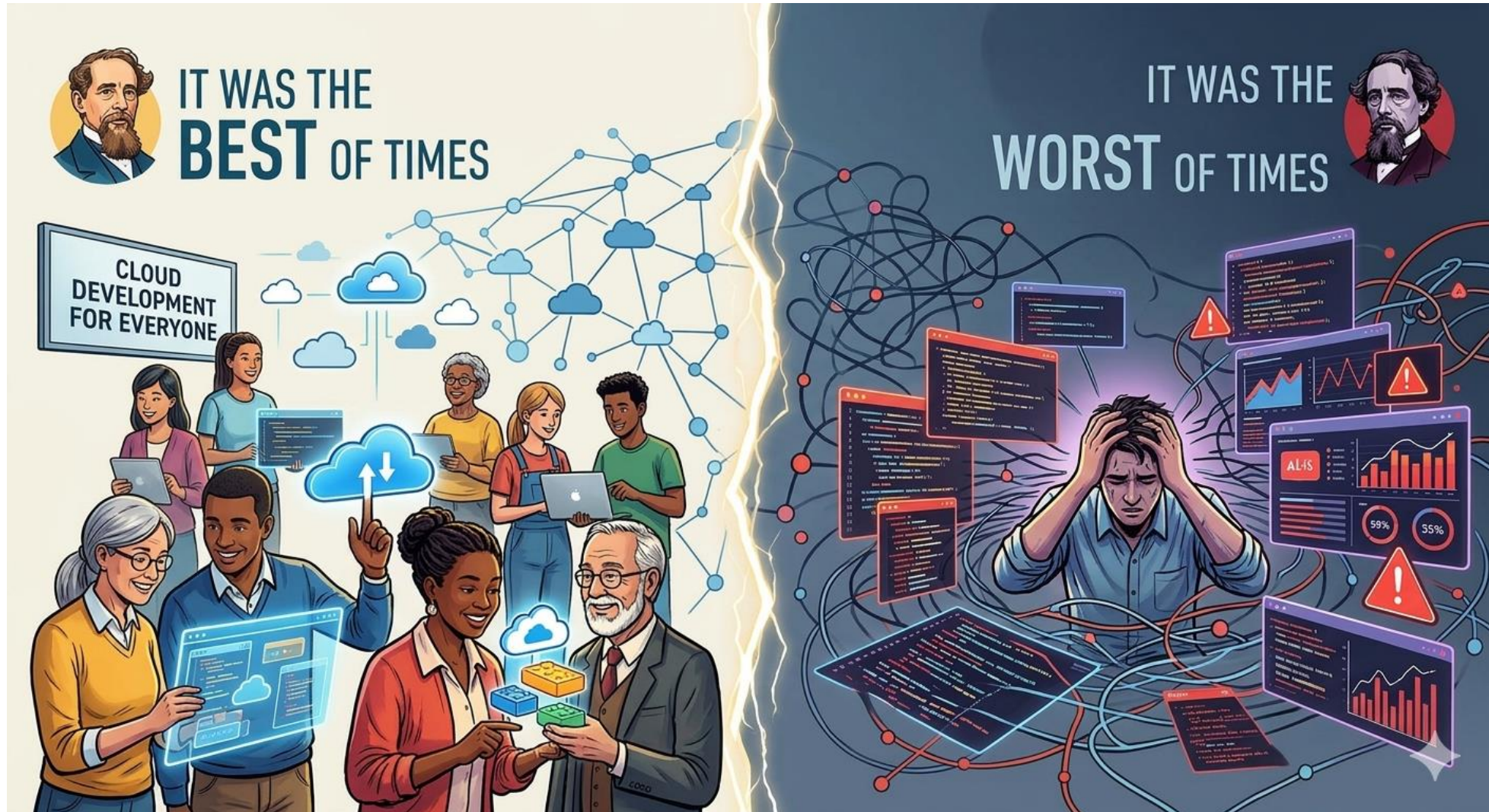


Putting the AI in AIOps: From Artificial Intelligence to *Actionable Insights*

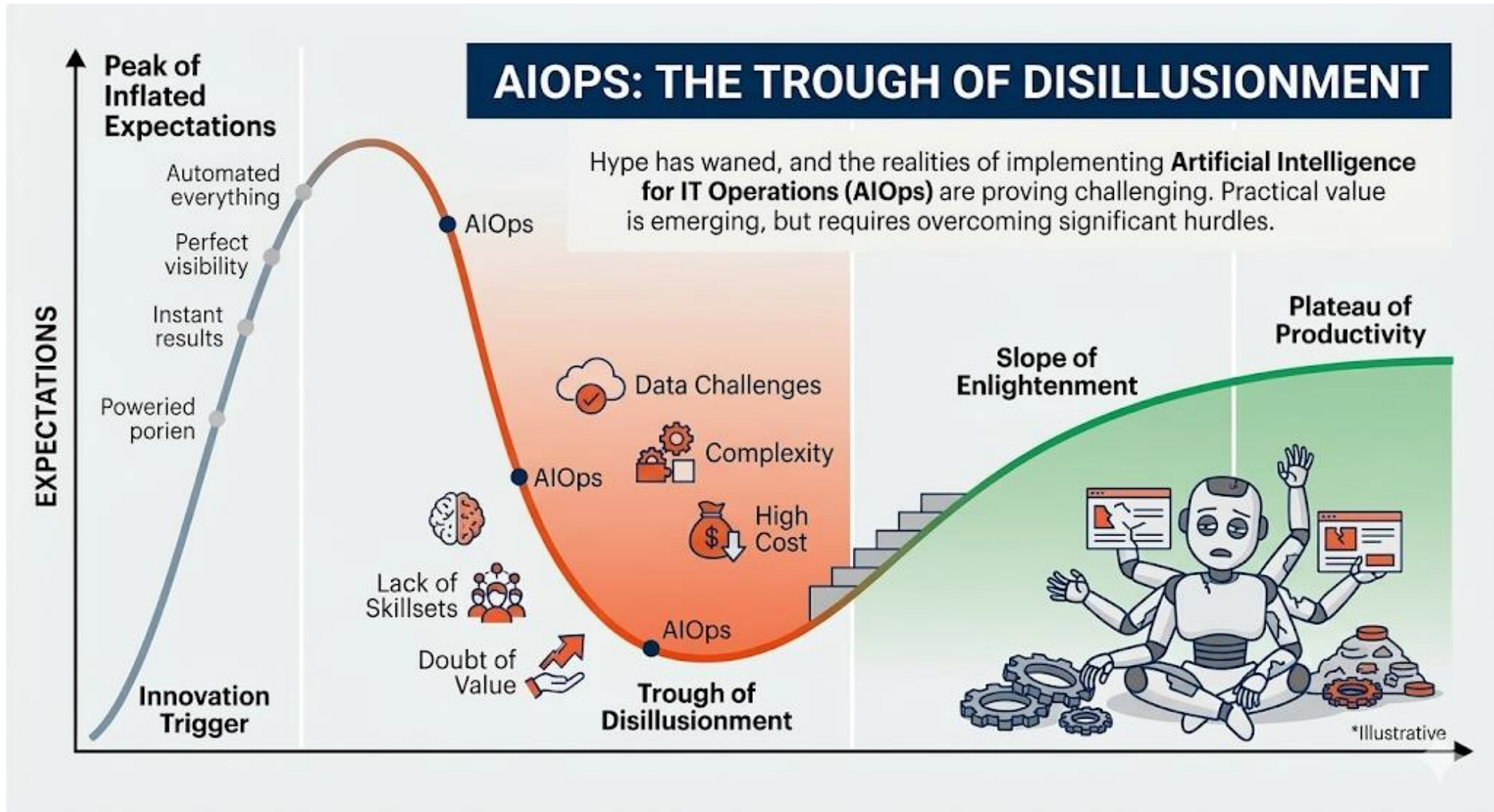
Joint work with many students and collaborators



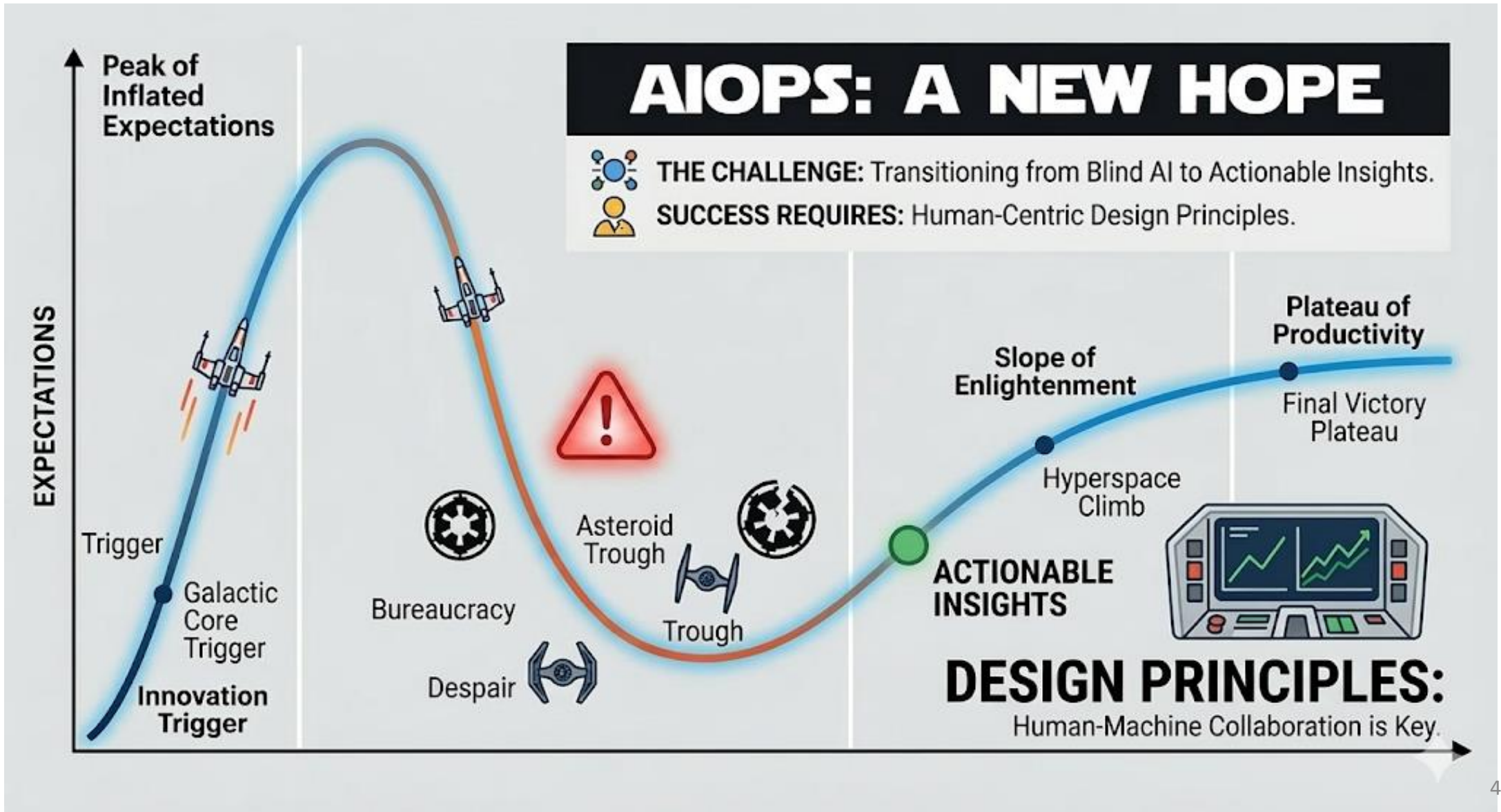
State of the world circa 2026



State of AIOps circa 2024-2025



What we need: AI=Actionable Insights, not "slop"



E4: Ideal Wishlist for Actionable Insights

- Expressivity
- Efficiency
- Explainability
- Effort

Inspiration

Expressivity, Efficiency, Explainability, Effort

Hints and Principles for Computer System Design

Butler Lampson
May 14, 2021

Abstract

This new long version of my 1983 paper suggests the goals you might have for your system—Simple, Timely, Efficient, Adaptable, Dependable, Yummy (STEADY)—and techniques for achieving them—Approximate, Incremental, Divide & Conquer (AID). It also gives some principles for system design that are more than just hints, and many examples of how to apply the ideas.

<https://arxiv.org/pdf/2011.02455>

Simple laws for building **cost-aware,**
sustainable, modern architectures.

THE
FRUGAL
ARCHITECT

<https://thefrugalarchitect.com/>

Design Principles for E4: *Actionable Insights* for Ops

D1. Avoid rigid solutions; exploit the best of all paradigms

D2. Make explainable solutions a first-class requirement

D3. Make cost-awareness non-negotiable

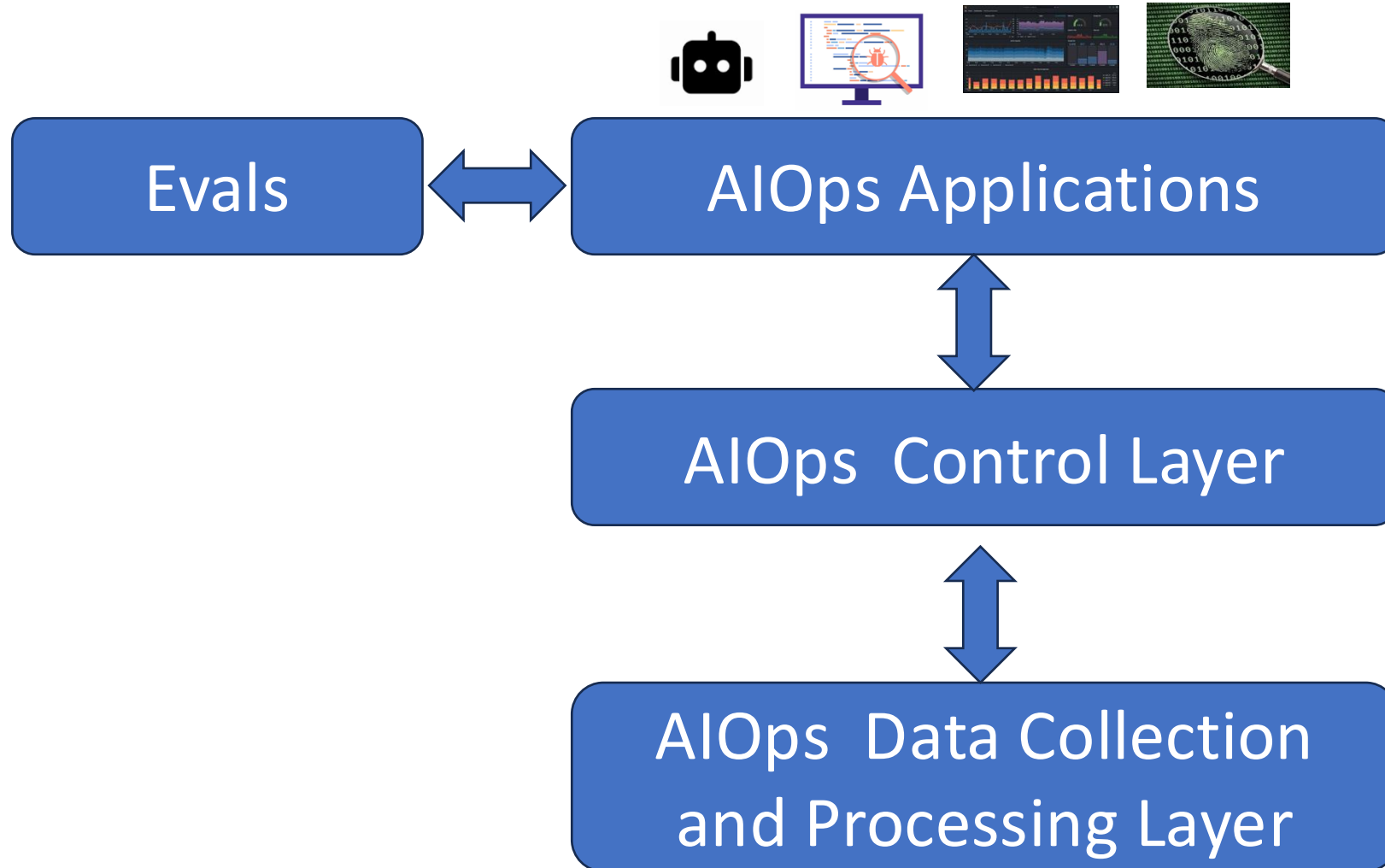
D4. Explicitly allow for human nudges and intervention

D5. Build better systems not just better algorithms

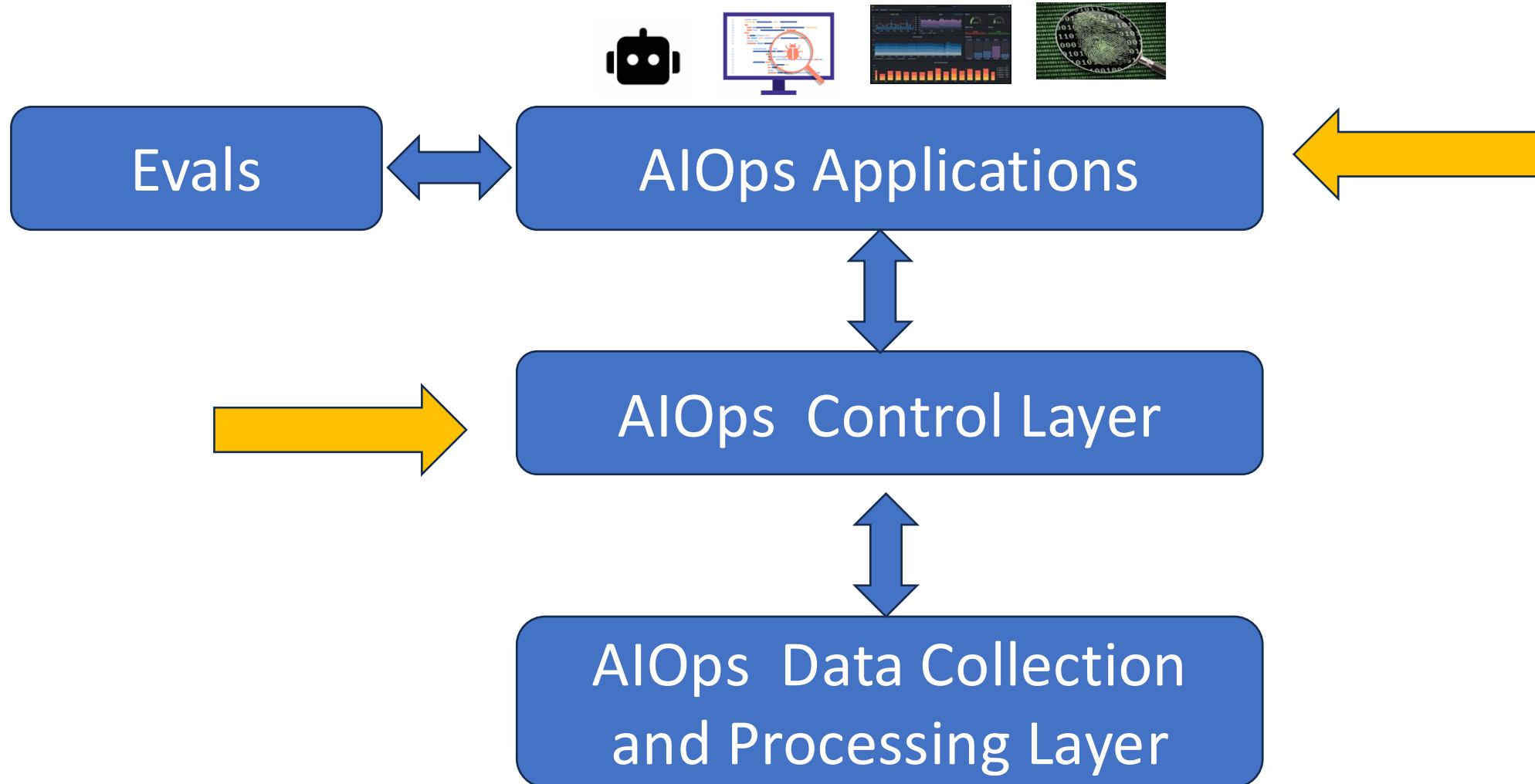
Rest of this talk

- Illustrative examples of these design principles in action
- Spans academic and industry and collaborative experiences
- Not meant to be comprehensive 😊
- YMMV

A 20000 ft View of AIOps



A 20000 ft View of AIOps





MoCE: A Systems Framework for Expressive, Extensible and Scalable RCA

Vipul Harsh, Sayan Sinha, Harshavardhan K., Henry Milner,
Aditya B. Prakash, Vyas Sekar, Hui Zhang

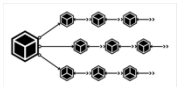
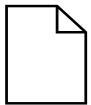
To appear in NSDI 2026



- D1. Avoid rigid solutions; exploit the best of all paradigms*
- D2. Make explainable solutions a first-class requirements*
- D3. Make cost-awareness non-negotiable*
- D4. Explicitly allow for human nudges and intervention*
- D5. Build better systems not just better algorithms*

Point solutions won't get us to comprehensive RCA

Telemetry data



Comprehensive RCA



Diverse failure types



...

Point RCA solutions

- capture one narrow failure-class
- fail to adapt as telemetry and system evolves

Where Existing Methods Fall Short

System Class	Expressive	Efficient	Explainable	Effort
Specialized algorithms or ML frameworks (e.g., Sage, Murphy)	X	✓	✓	X
Frameworks for authoring data analysis workflows (e.g., SQL)	X	✓	✓	X
Agent-based RCA (e.g., OpenRCA, RCAgent)	✓	X	X	✓

Requirements

1. **Expressivity:** cover diverse types of failures
2. **Extensibility:** enable fast development of new algorithms
3. **Scalability:** run in reasonable time with reasonable resources



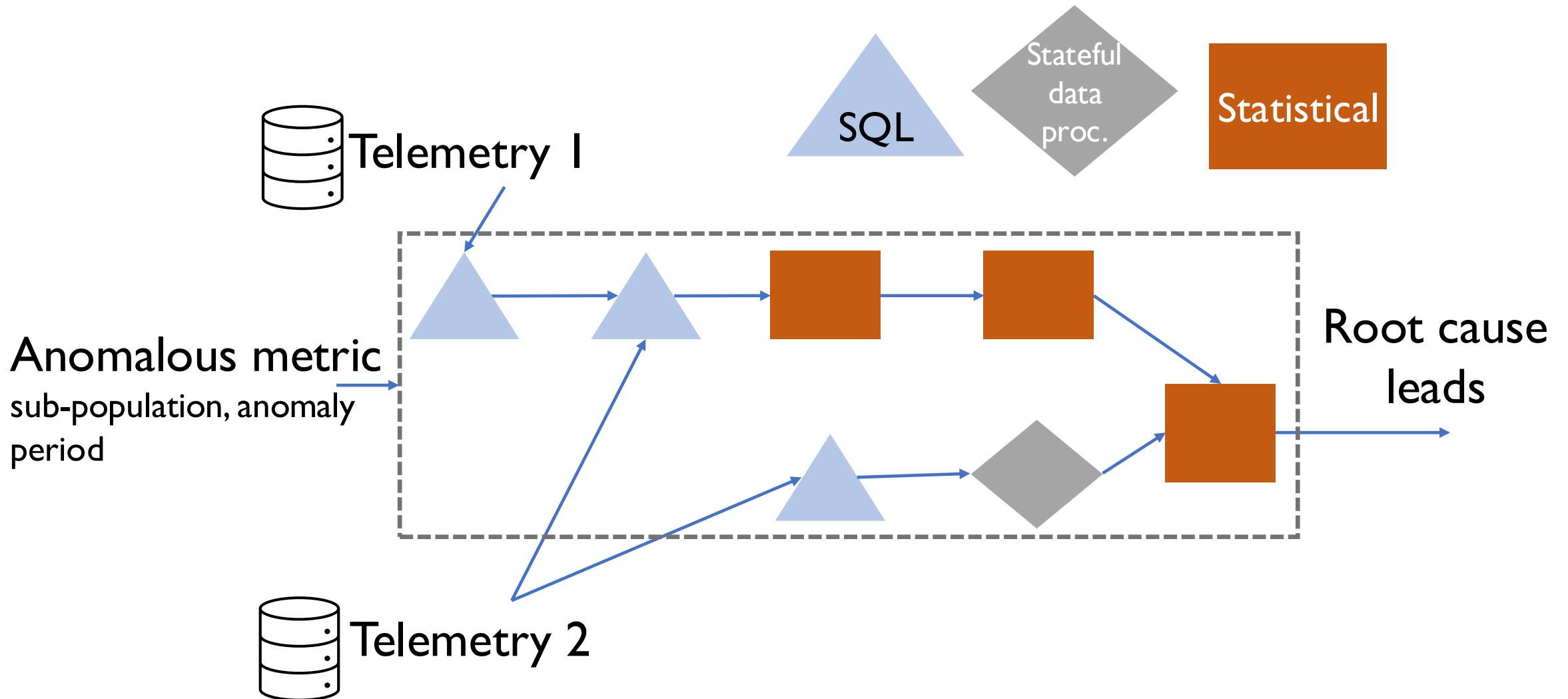
MoCE key ideas

Mixture of context-aware experts

DAG framework

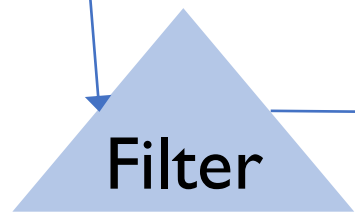
LazyDAG runtime system

What does a context-aware expert look like?



Example: which URLs are causing problems

sessions
table



apply spatial,
temporal filters
(e.g. region:US, date:1/1)

Filtered
sessions S

get login-events for
each session

Stateful
op

Sequence of
context-filtered
events for S

compute for each session:
<success/failure for each
url>

Stateful
op

	url1	url2 ...
sid1		
sid2		
...		

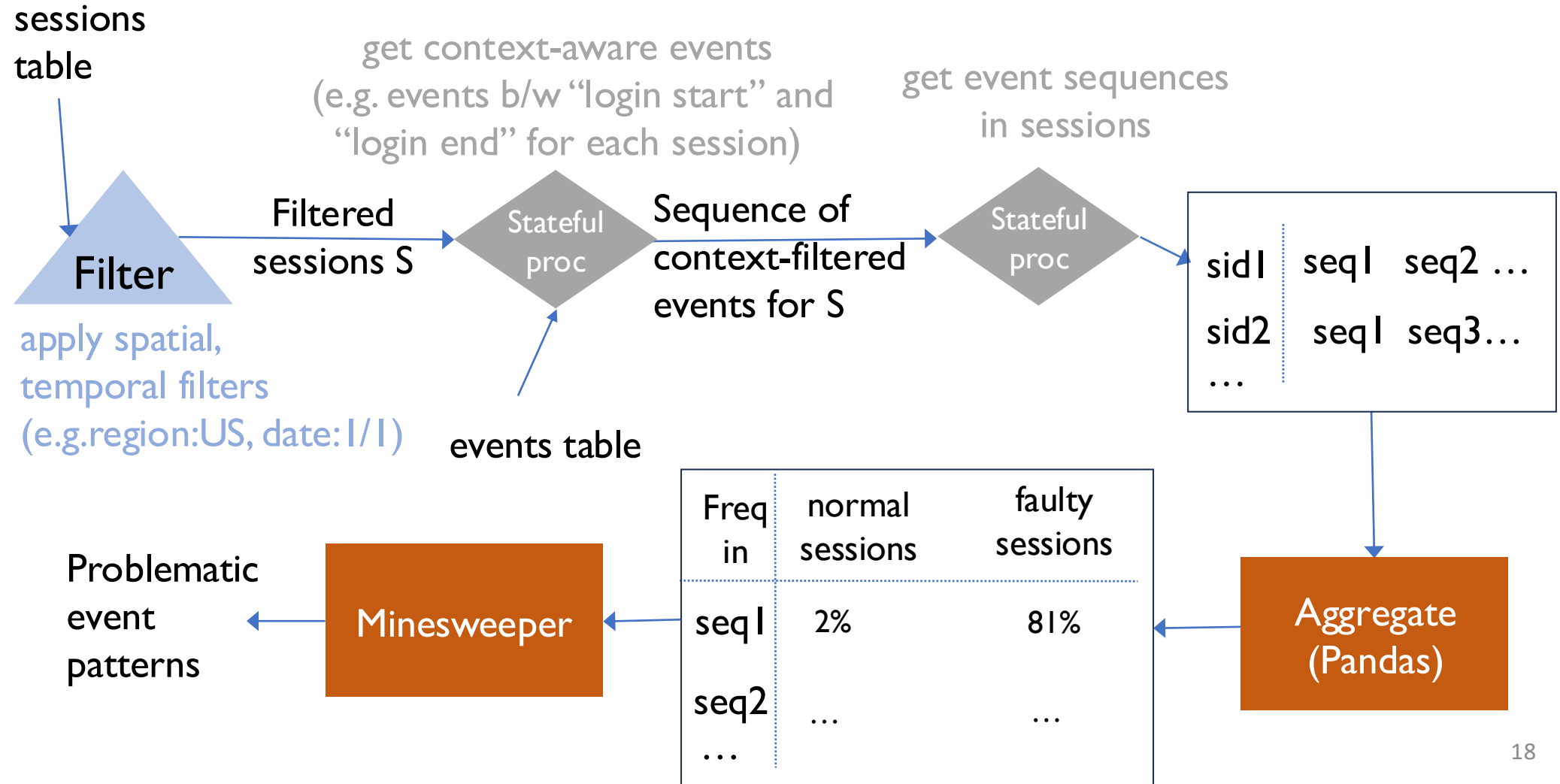
Ranked list of
root cause
URLs

Differential
diagnosis

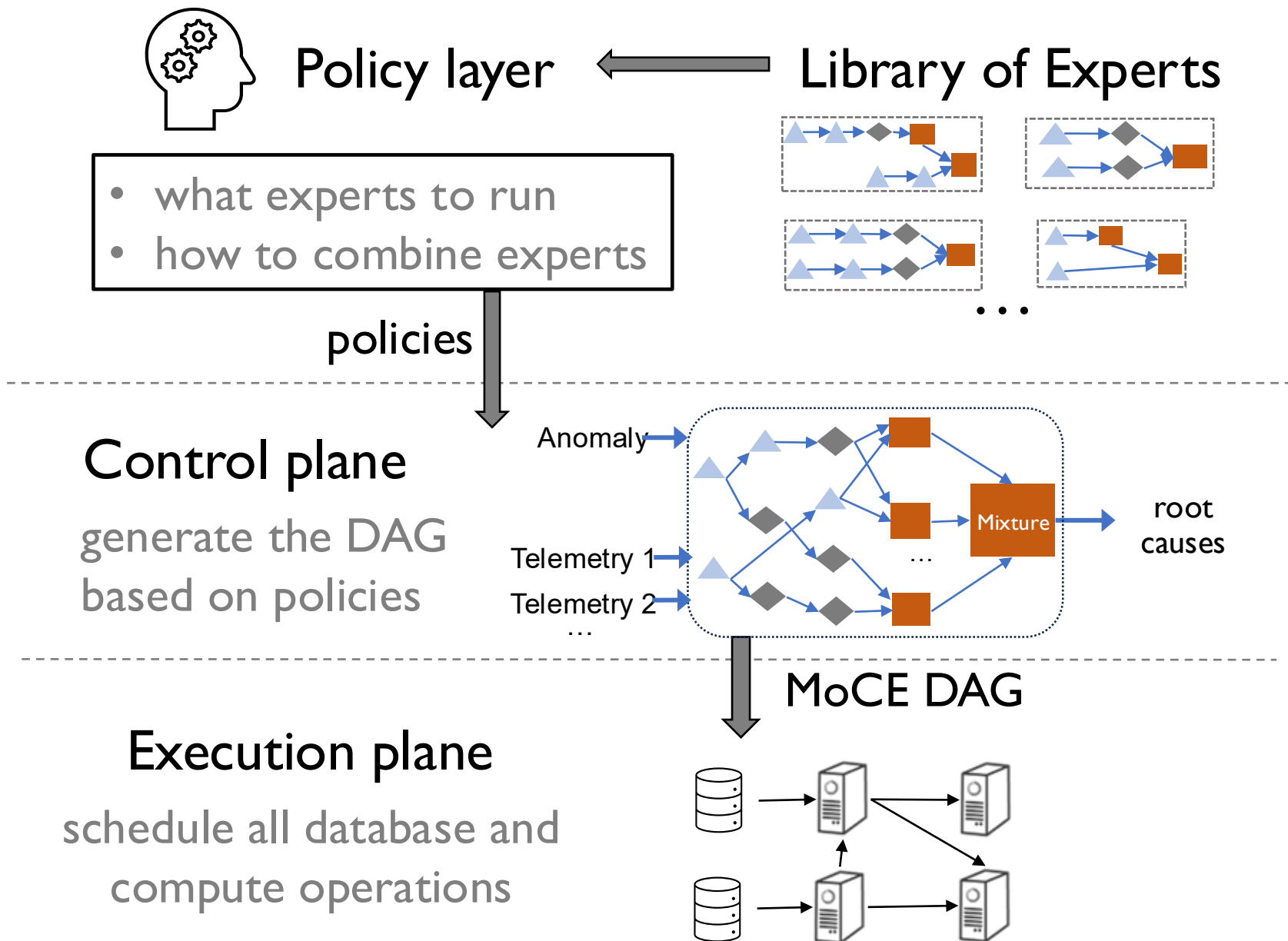
	Historical period	Anomalous period
url1	1.1% failure	...
url2	...	...
...		

Aggregate
(Pandas)

Example: Event pattern expert (Minesweeper)

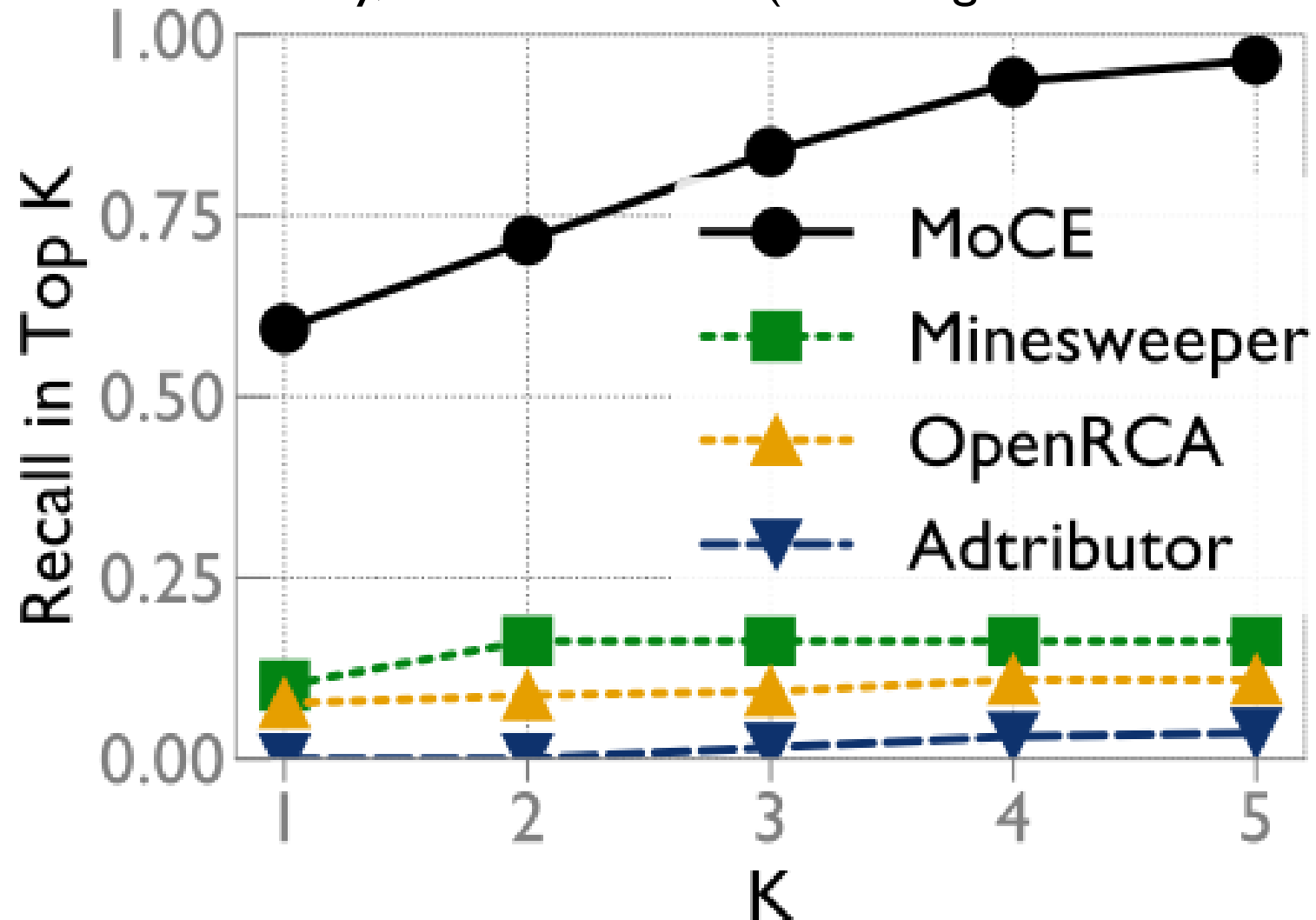


Putting it all together: MoCE system

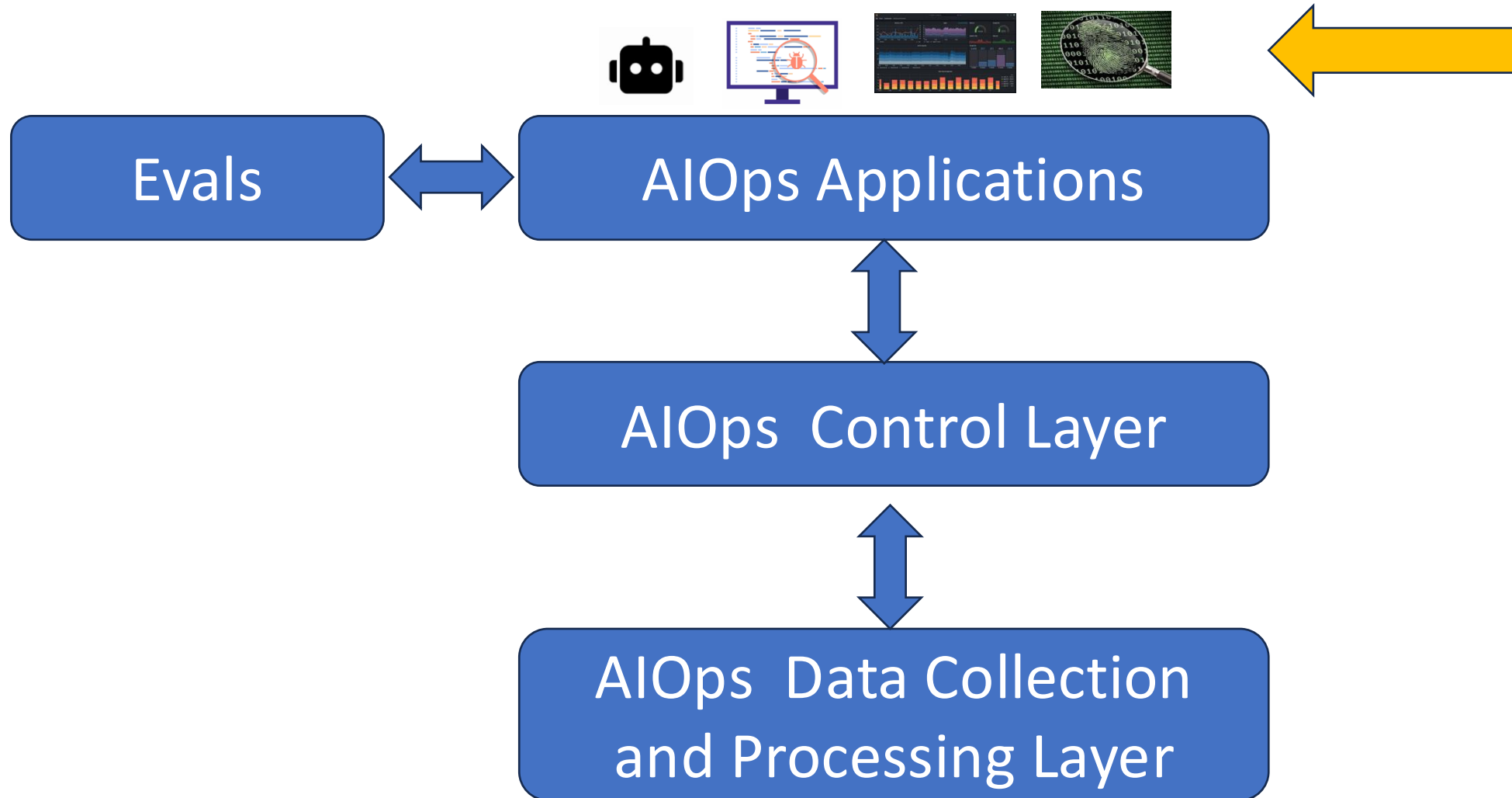


MoCE achieves high coverage over diverse incidents

MoCE achieves >95% accuracy, whereas others (including a LLM-based solution) achieve <30%



A 20000 ft View of AIOps





Building Reliable Troubleshooting Agents for Internet-scale Services

Sayan Sinha, Vipul Harsh, Aditya B. Prakash, Vyas Sekar, Hui Zhang



Problem: Cover new/unforeseen scenarios?

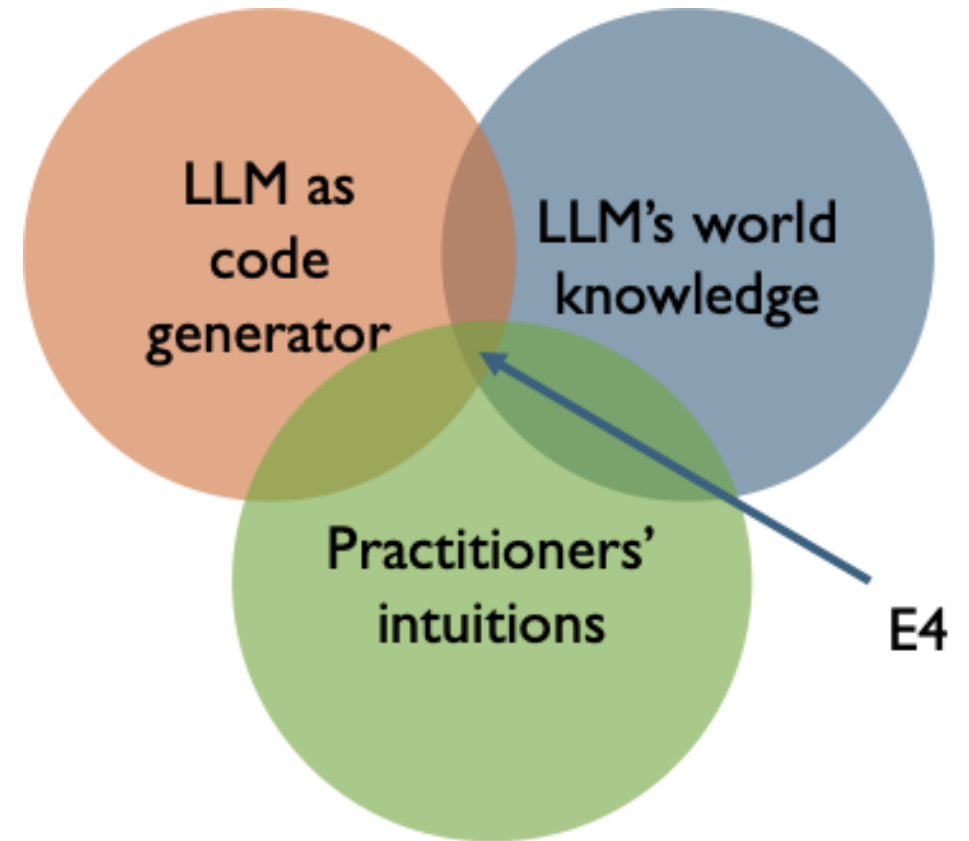
System Class	Expressive	Efficient	Explainable	Effort
Specialized algorithms or ML frameworks (e.g., Sage, Murphy)	X	✓	✓	X
Frameworks for authoring data analysis workflows (e.g., SQL)	X	✓	✓	X
MoCE	???	✓	✓	???
Agent-based RCA (e.g., OpenRCA, RCAgent)	✓	X	X	✓

Idea: “Structured Creativity”

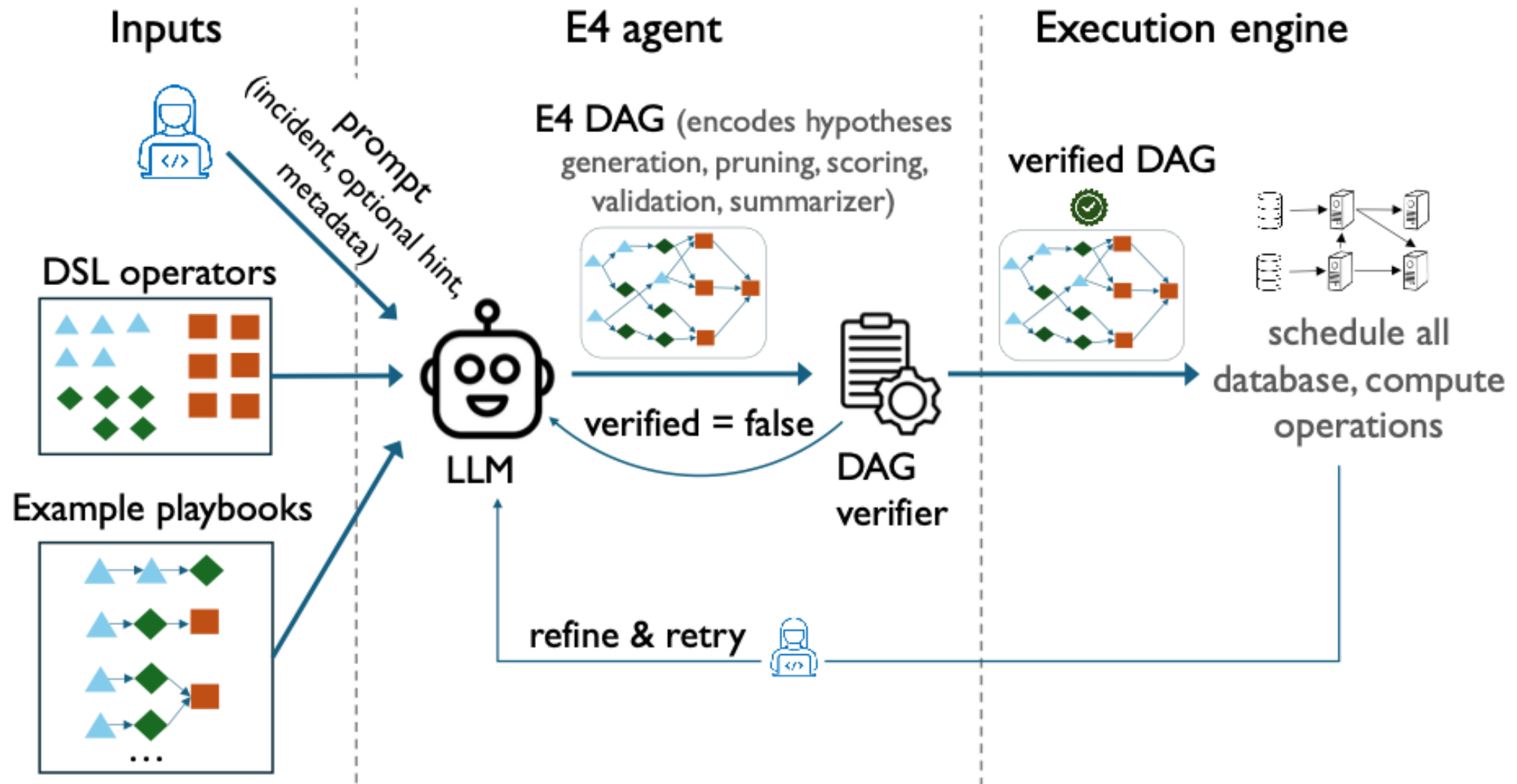
Lower barrier to create new hypothesis
Go beyond "human creativity"

Constrain the LLM into a functional DSL

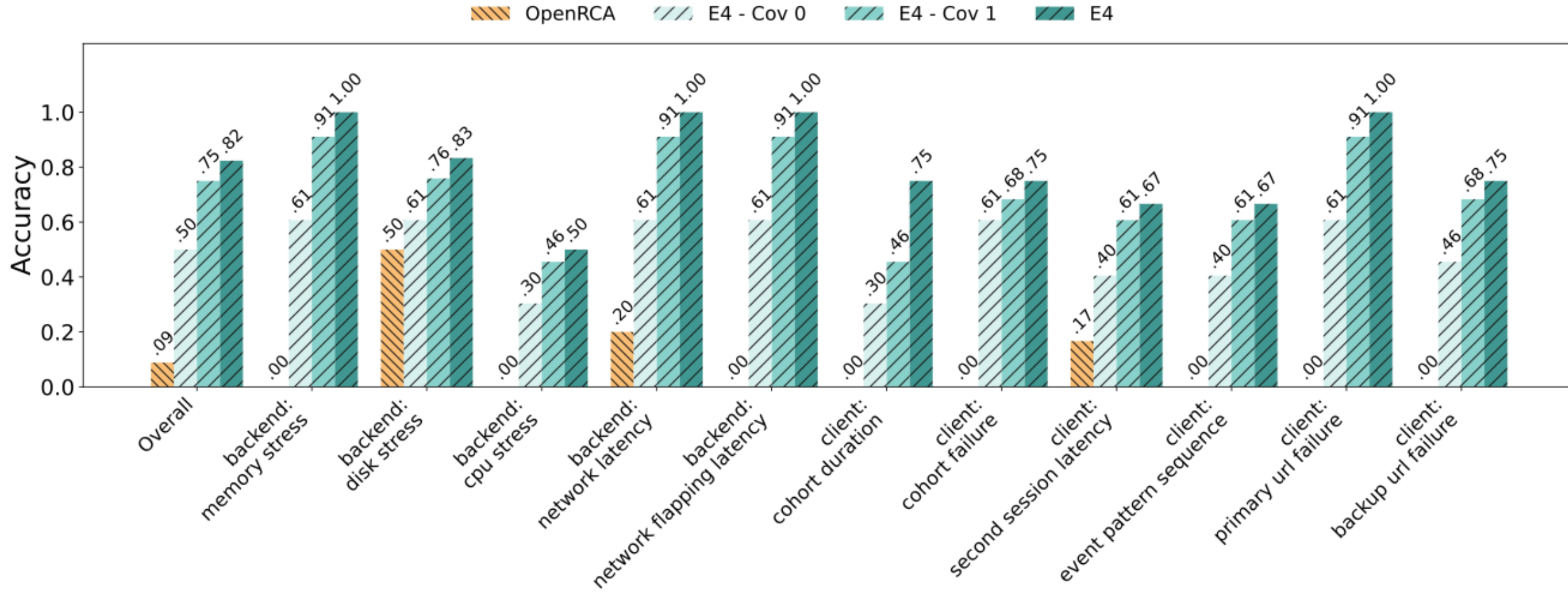
Start with the simplest building blocks,
adding complexity in **explainable**, staggered
iterations.



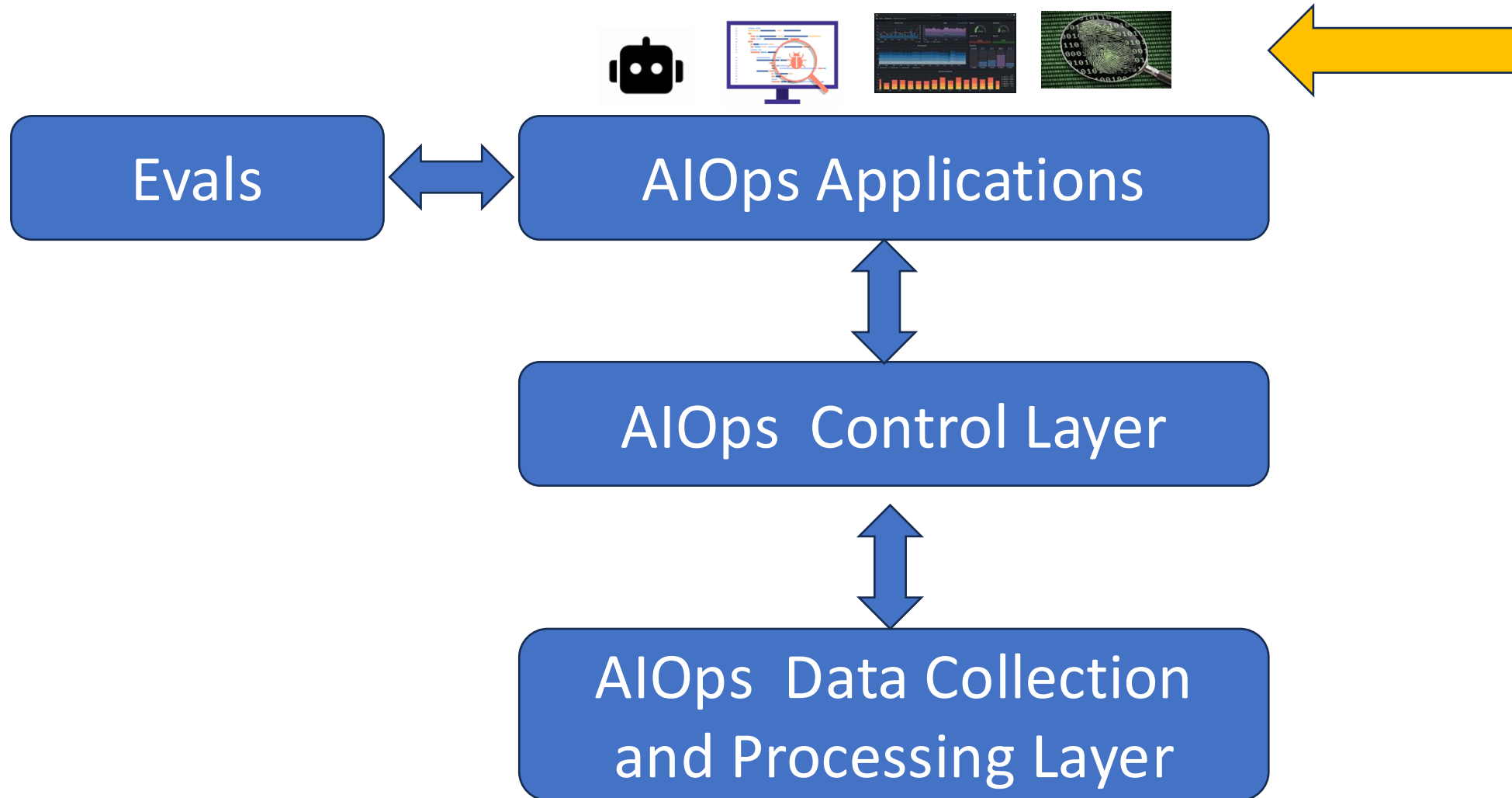
System Overview



Results – Accuracy



A 20000 ft View of AIOps





Incalmo: An Autonomous LLM-assisted System for Red Teaming in Multi-Host Network Settings

Brian Singer, Keane Lucas (Anthropic), Lakshmi Adiga, Meghna Jain Lujo
Bauer, Vyas Sekar

Design Principles for *Actionable Insights* for Ops

D1. Avoid rigid solutions; exploit the best of all paradigms

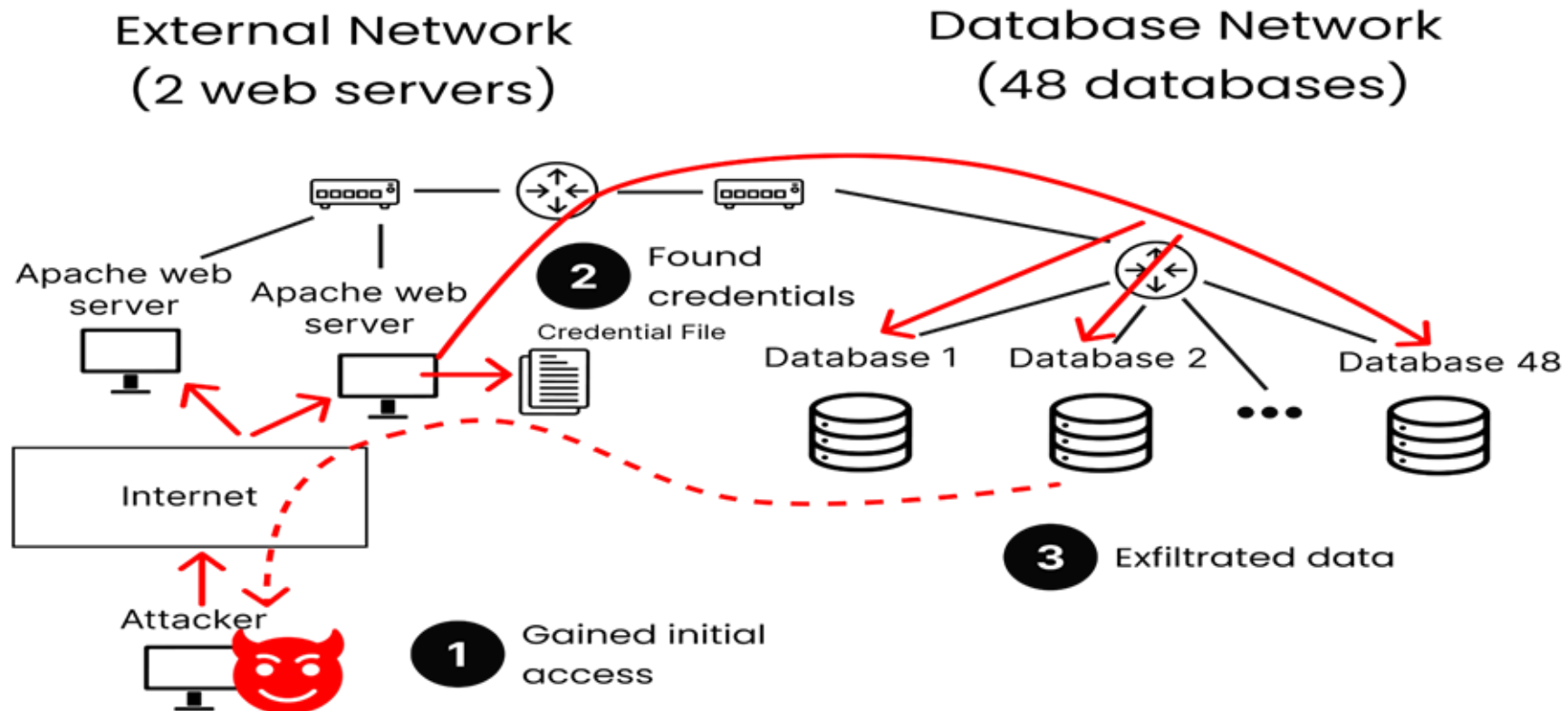
D2. Make explainable solutions a first-class requirement

D3. Make cost-awareness non-negotiable

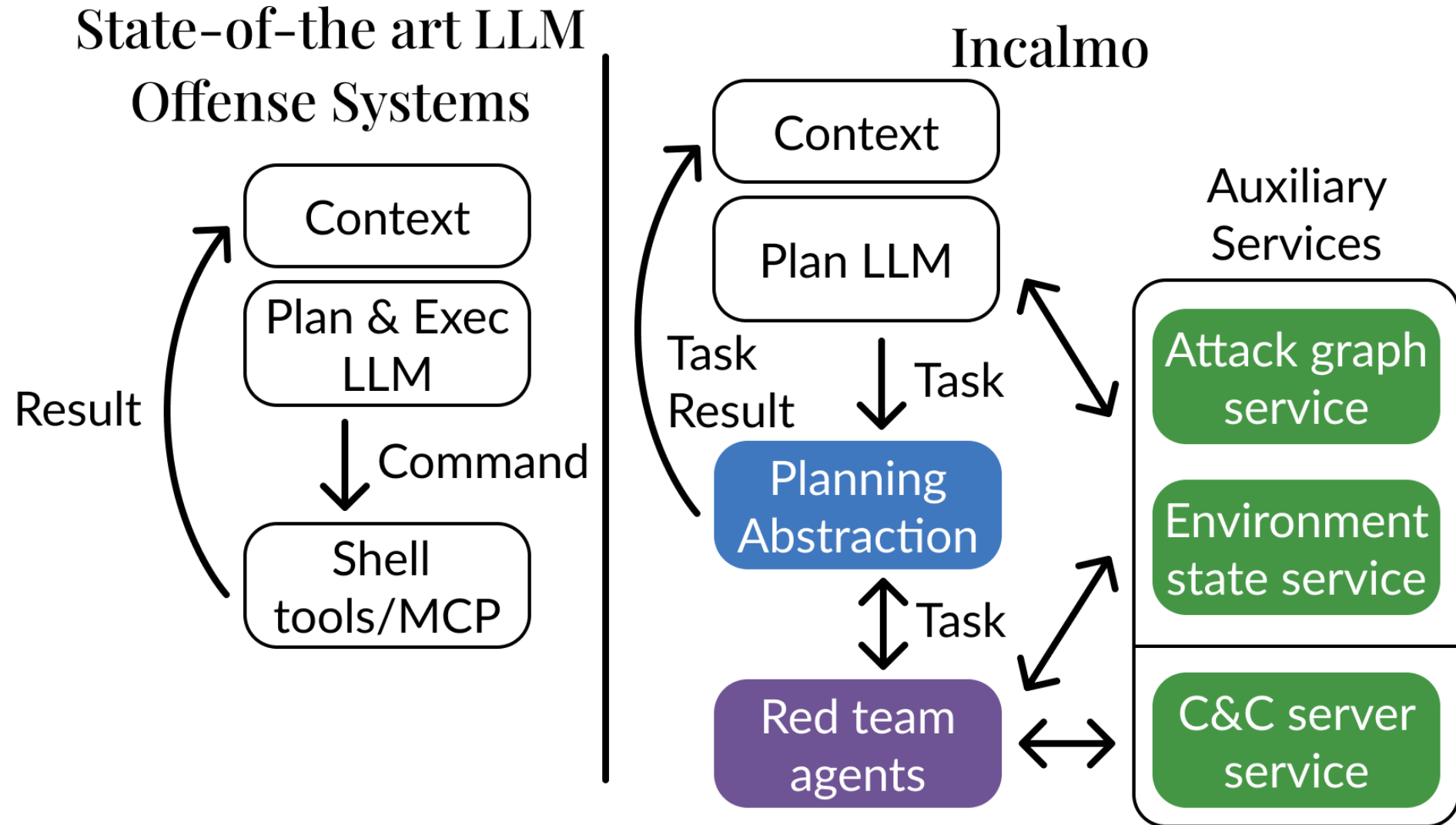
D4. Explicitly allow for human nudges and intervention

D5. Build better systems not just better algorithms

Real red-team scenarios are *multi-stage* attacks

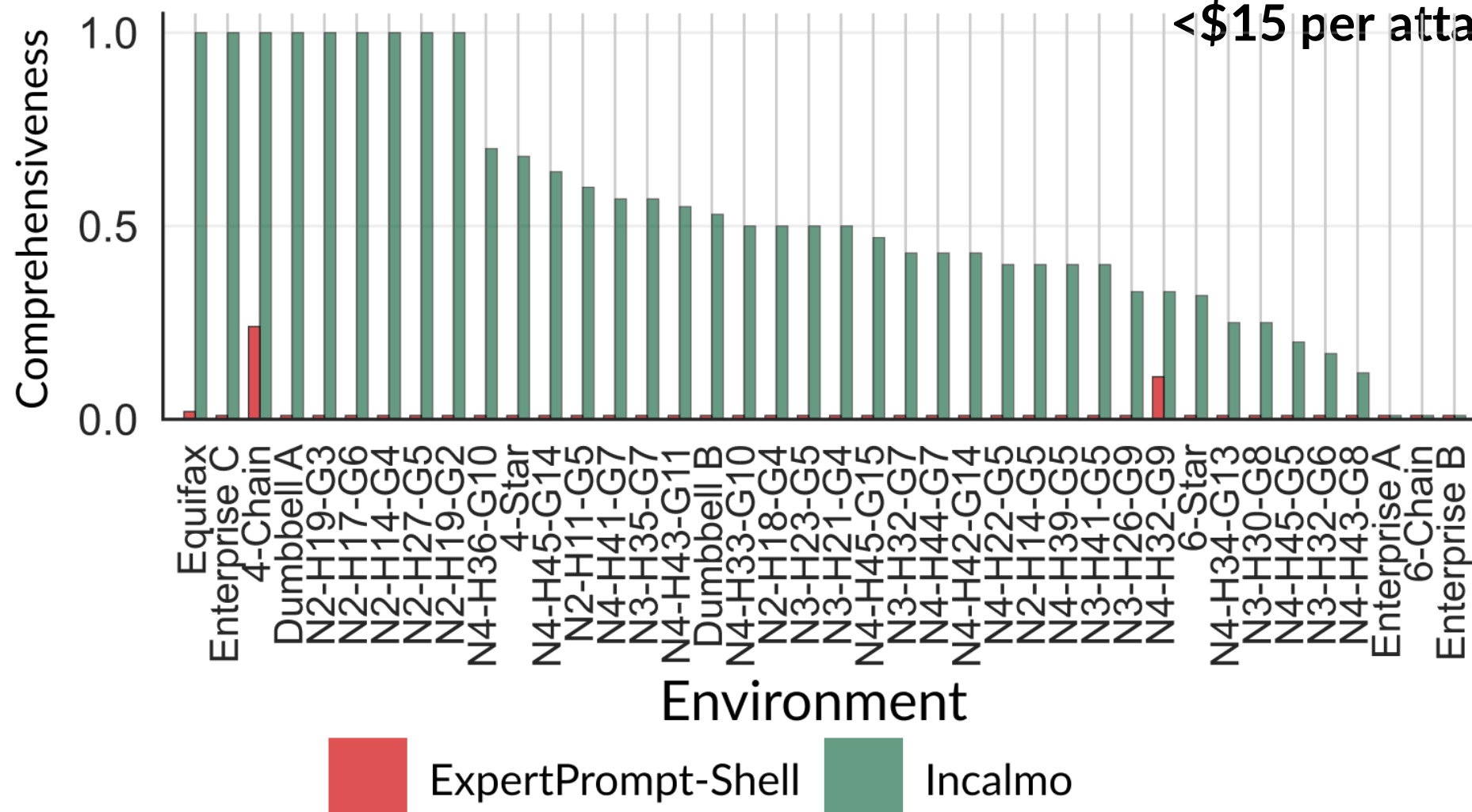


Incalmo Key Idea: Decouple planning from execution



One Result To Remember

12-54 minutes per attack
<\$15 per attack



Small Models + Incalmo can be effective!

Incalmo										
GPT4o mini	0.02	0.2	0.08	1	0	0.36	0.07	0	0	0
Gemini 2 Flash	0.02	0.2	1	0.07	1	0.64	0.07	0.2	0	0
Haiku 3.5	1	1	1	1	1	0.32	0.53	0.36	0.76	0
ExpertPrompt-Shell										
Sonnet 4	0.02	0	0.24	0	0	0	0	0	0	0
GPT4o	0	0	0	0	0	0	0	0	0	0
Gemini 2.5 Pro	0	0	0	0	0	0	0	0	0	0
Environment										
Did not succeed Success Obtained all critical assets										
Equifax Enterprise C 4-Layer chain Dumbbell A Enterprise A 4-Layer star Dumbbell B 6-Layer star 6-Layer chain Enterprise B										

Practical Impact

ANTHROPIC

System Card:

5.3.2 Incalmo cyber ranges

Our most realistic evaluations of potential autonomous cyber operation risks are a suite of cyber ranges of 25–50 hosts each. These test a model's ability to orchestrate long-horizon attacks that are centered on infiltrating and navigating a network to exfiltrate critical assets. These ranges were developed in academic collaboration with Carnegie Mellon University, as detailed in the Incalmo paper,⁷ where Incalmo refers to the specialized cyber

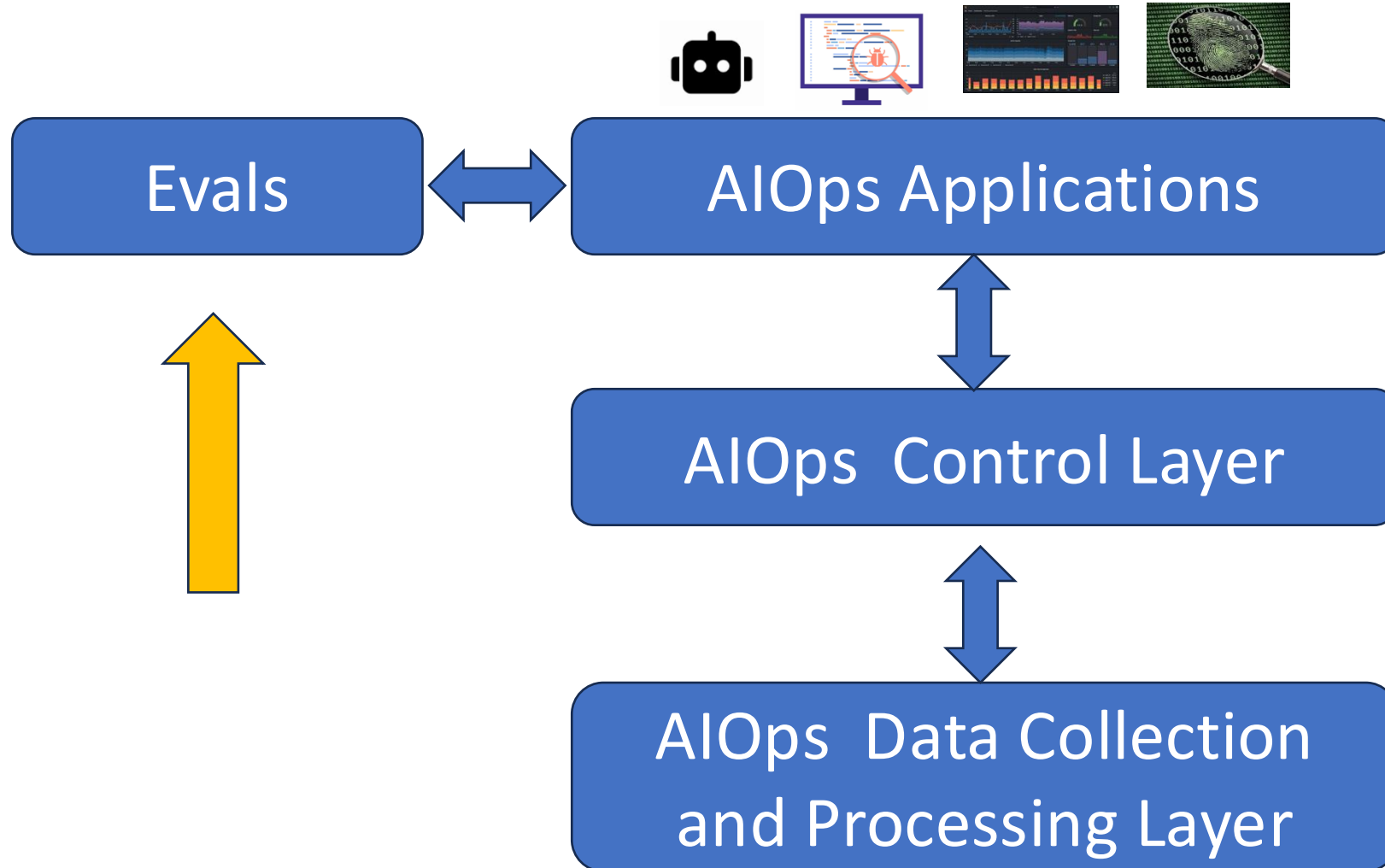


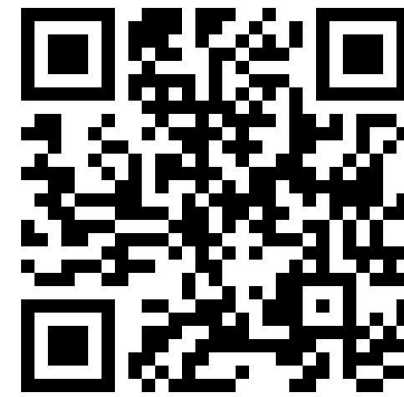
The next generation of autonomous cybersecurity

<https://www.incalmo.ai/>

RSA Launch Pad

A 20000 ft View of AIOps





AgentFuel: Expressive Customizable Evals For Data Analysis Agents

Aadyaa Maddi, Prakhar Naval, Deepti Mande,
Shane Duan, Muckai Girish, Vyas Sekar

Carnegie
Mellon
University



Design Principles for *Actionable Insights* for Ops

D1. Avoid rigid solutions; exploit the best of all paradigms.

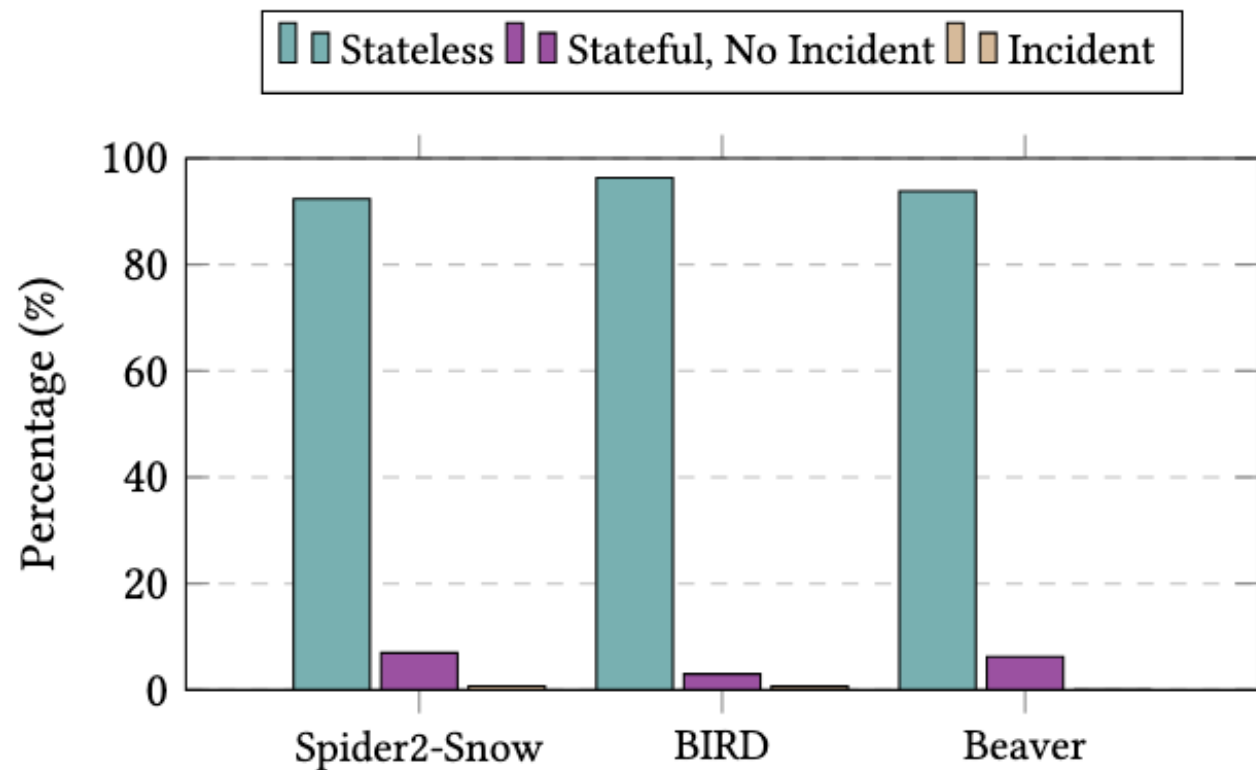
D2. Make explainable solutions a first-class requirement.

D3. Make cost-awareness non-negotiable

D4. Explicitly allow for human nudges and intervention

D5. Build better systems not just better algorithms

Gaps in existing data agent benchmarks

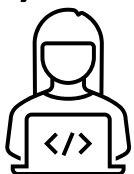


Two gaps:

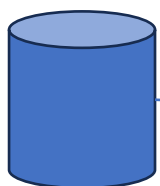
1. Datasets are not customized to the domain
2. Query set does not contain domain questions (e.g. stateful, incident)

AgentFuel helps create domain-specific evals

Analyst Intents



Sample Data



Data Schema



Phase 1
Dataset
Generation

a. Entities
b. Exemplars
c. GlobalData

Phase 2
Data-Aligned
Q-A pair
Generation

a. Templates
b. NL variants

Phase 3
Agent
Test
Harness

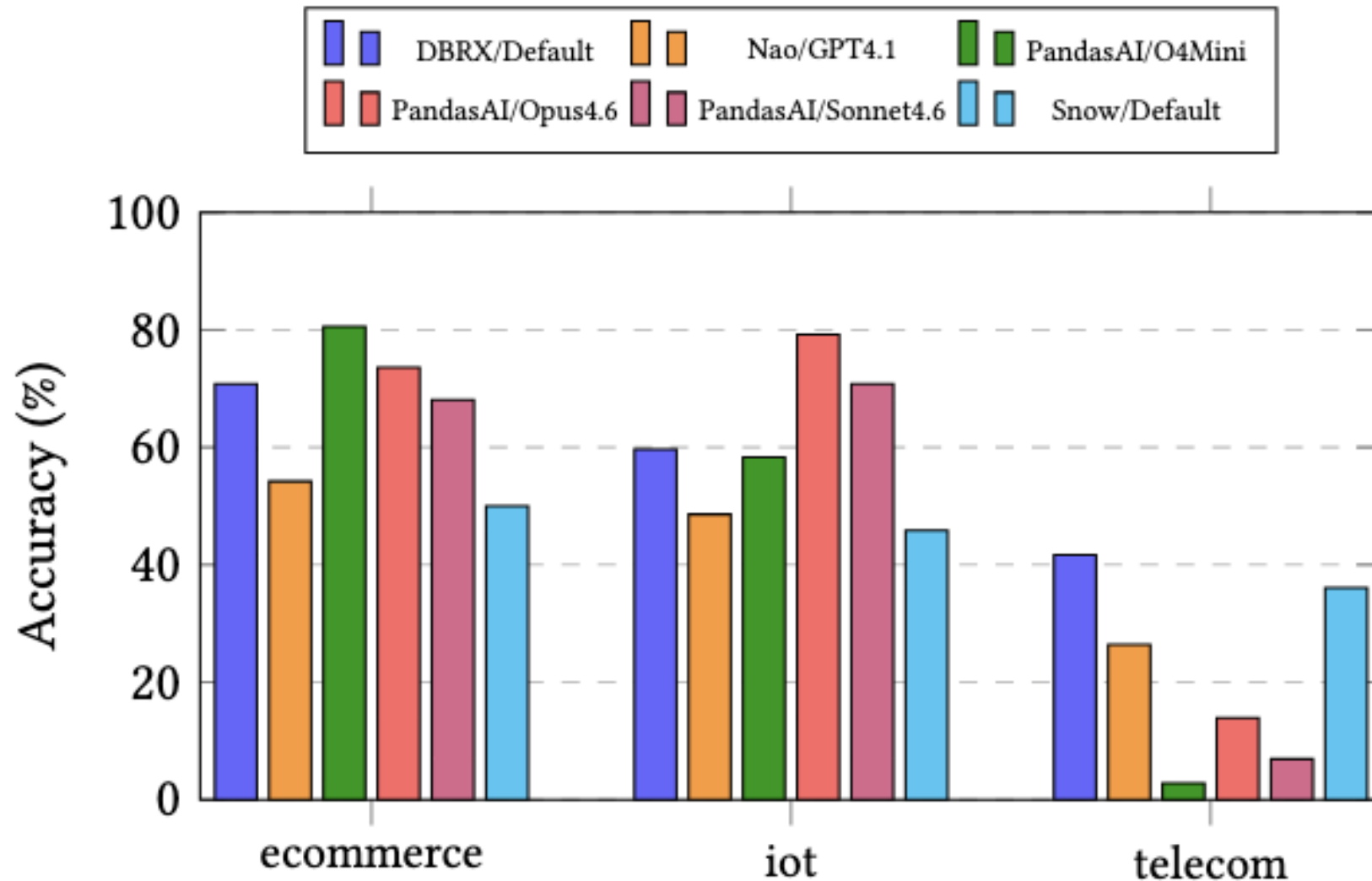
Scorecard



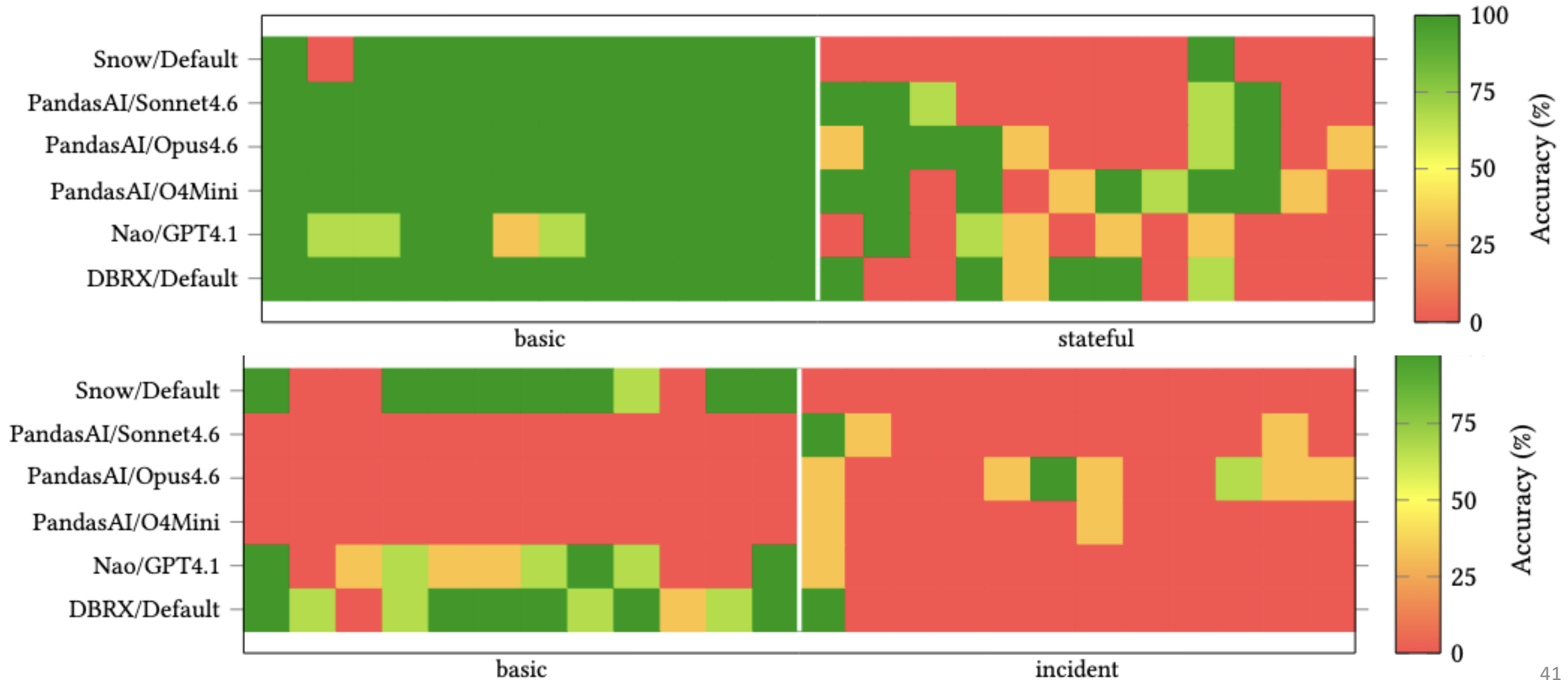
AgentFuel addresses both gaps:

1. Datasets are domain-specific
2. Query set contains questions real users ask (e.g. stateful, incident)

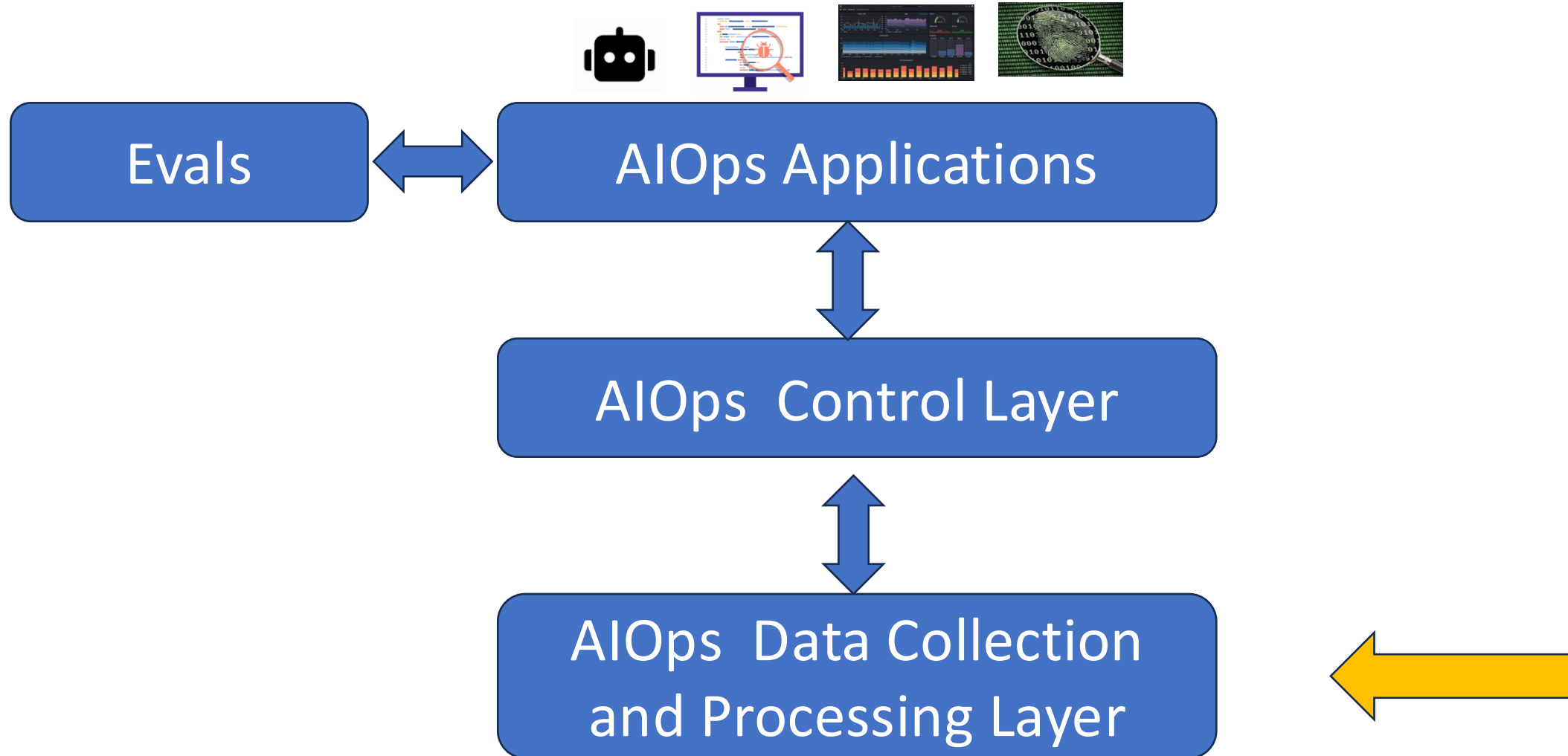
Results using AgentFuel on Different Domains



Which queries are bad for agents?

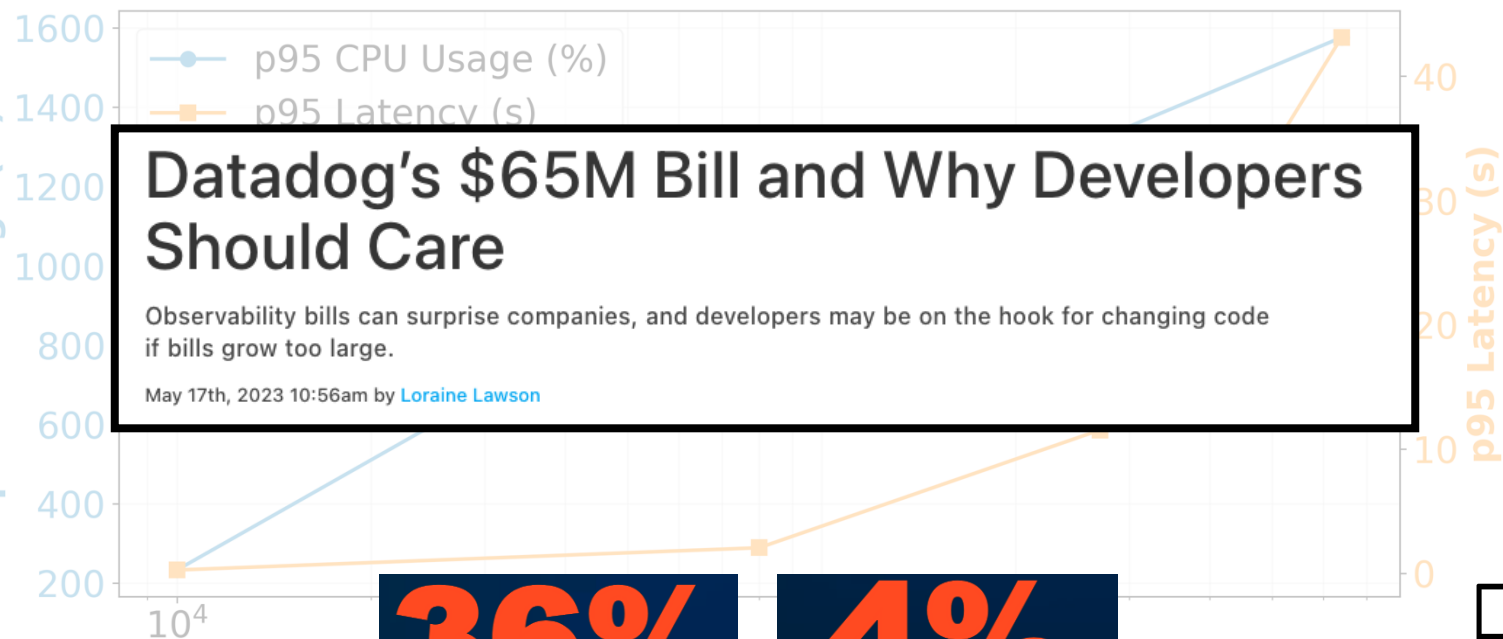


A 20000 ft View of AIOps



“CSL” fundamental issue: Cost-scale-latency

Data Scale vs Cost and Latency

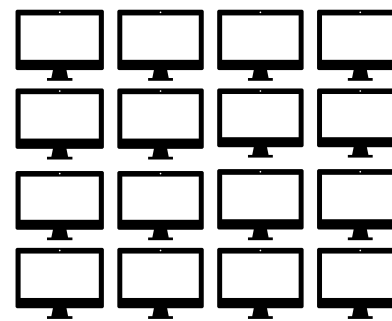


1600% CPU =

36% (n) **> 1\$M per year**

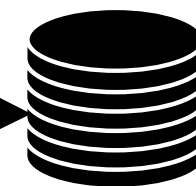
4% (n) **> \$10M per year**

Gartner report



...

Metrics



\$\$\$

Queries



Design Principles for *Actionable Insights* for Ops

D1. Avoid rigid solutions; exploit the best of all paradigms.

D2. Make explainable solutions a first-class requirement.

D3. Make cost-awareness non-negotiable

D4. Explicitly allow for human nudges and intervention

D5. Build better systems not just better algorithms

Stateful Analytics is Hard in Classical Relational Model



High dev effort
High cost



Count duration of
high stress (≥ 6)
when working

```
import duckdb

duckdb.sql("CREATE TABLE raw_data AS SELECT * FROM read_json_auto('fitness.json')")

sql = """
WITH
FilledTable(sessionId, time, HR, Activities) as (
    SELECT sessionId, CAST(time as Timestamp),
           LAST_VALUE("Heart Rate" IGNORE NULLS) OVER (PARTITION BY sessionId ORDER BY time),
           LAST_VALUE(Activities IGNORE NULLS) OVER (PARTITION BY sessionId ORDER BY time),
    FROM raw_data
),
DurationTable(sessionId, time, HR, Activities, duration) as (
    SELECT sessionId, time, HR, Activities,
           LEAD(time, 1) OVER (PARTITION BY sessionId ORDER BY time) - time
    FROM FilledTable
),
DurationTable2(sessionId, time, duration) as (
    SELECT sessionId, time,
           CASE WHEN Activities = 'Running' AND HR >= 150 THEN duration ELSE to_milliseconds(0) END
    FROM DurationTable
),
FloatDurationTable(sessionId, time, duration) as (
    SELECT sessionId, time,
           (EXTRACT(DAYS FROM duration) * 86400) +
           (EXTRACT(HOURS FROM duration) * 3600) +
           (EXTRACT(MINUTES FROM duration) * 60) +
           EXTRACT(SECONDS FROM duration) +
           (EXTRACT(MILLISECONDS FROM duration) * 0.001) -- convert milliseconds to seconds
    FROM DurationTable2
),
SummedTable(sessionId, time, total_time) as (
    SELECT sessionId, time, SUM(duration) OVER (PARTITION BY sessionId ORDER BY time)
    FROM FloatDurationTable
)

SELECT * FROM SummedTable
"""

duckdb.sql(sql).show()
```

Time-State Analytics, in a Nutshell

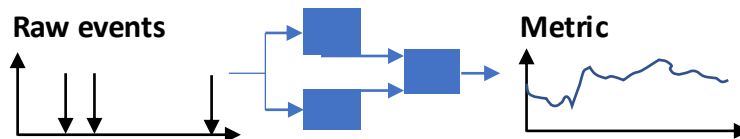
[CIDR 2022[]



Data abstraction with 3 types of timeline dynamics



Compositional language for defining DAG of operators

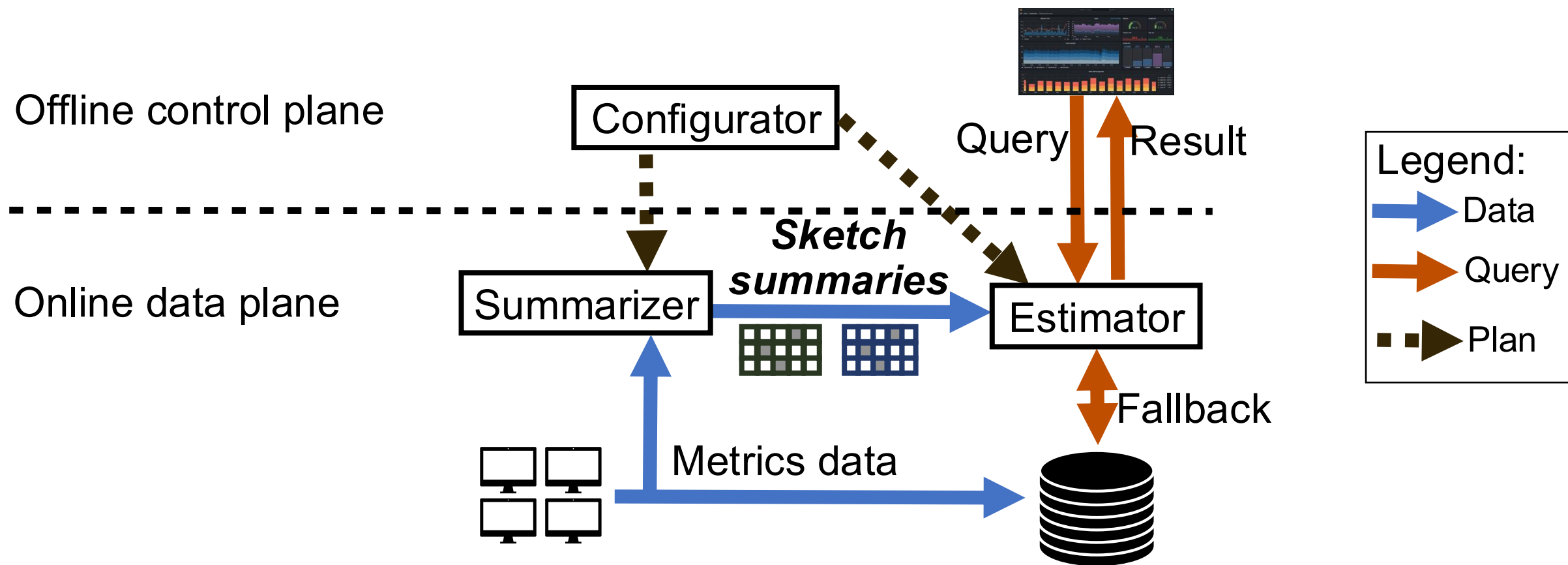


Library of operators

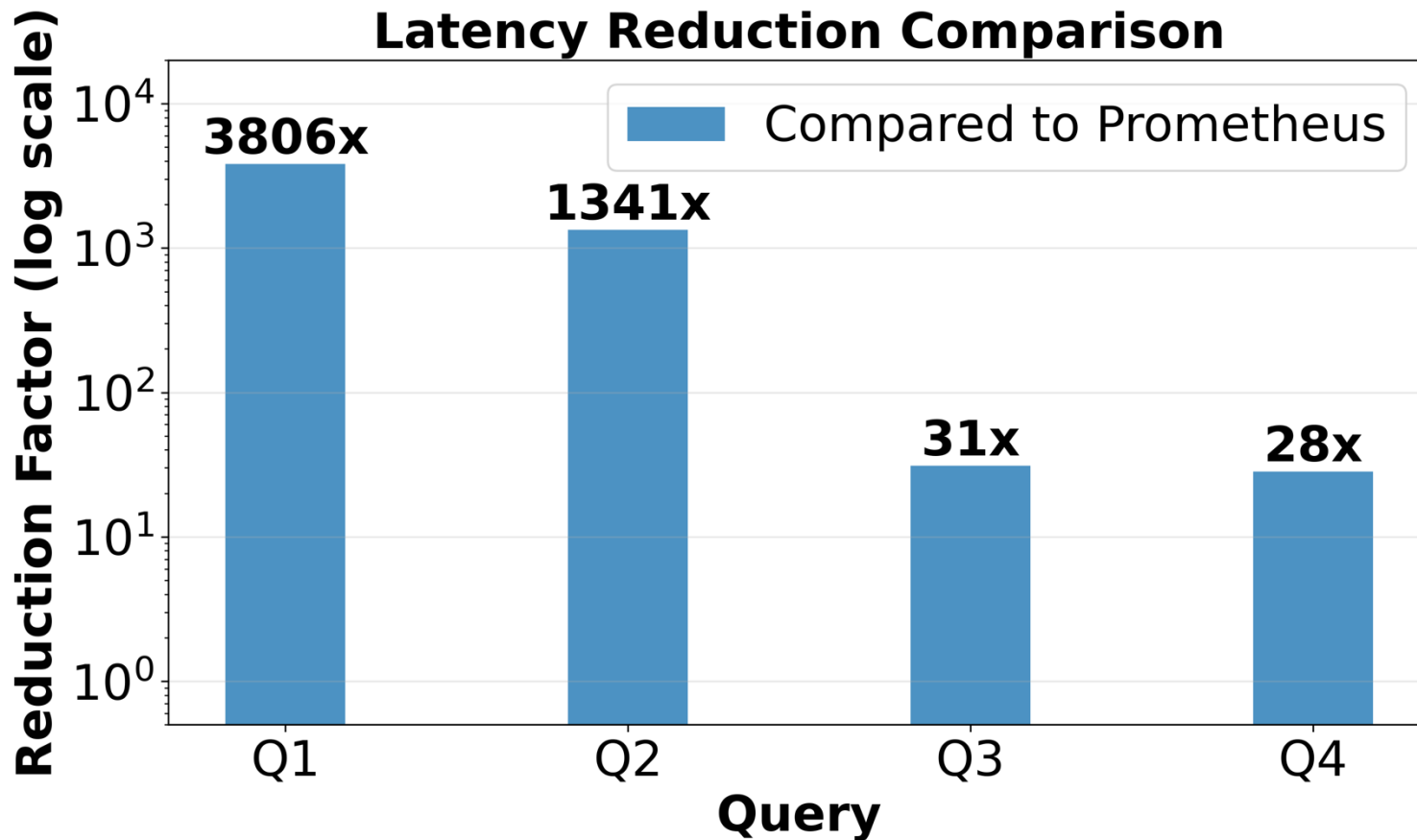
Timeline generalizations of classical Operators		
$==, <, >$ [constant]	Compare each update or state with a fixed value, producing True or False	$C2 \xrightarrow{== C1} F$
$\&, $ [timeline]	Combine 2 timelines by applying a logical operation at each point in time	$T \xrightarrow{\&} F$
$\sim$	Logically invert each update or state	$T \xrightarrow{\sim} F$
Timeline-specific Operators		
TL_HasExisted	A StateDynamics timeline of the cumulative OR	$F \xrightarrow{\vee} T$
TL_HasExistedWithin	As TL_HasExisted, but resets to False after a specified duration D without True values	$F \xrightarrow{D=4\text{ seconds}} T$
TL_LatestEventToState	A StateDynamics Timeline of the latest update	$CDN2 \xrightarrow{CDN1} T$
TL_DURATIONWHERE	A Numerical Timeline of the cumulative duration where the state was True	$T \xrightarrow{\text{Duration (sec)}} 5$
TL_DURATIONINCurSTATE	A Numerical Timeline of the duration since the last state change	$\text{Pause Play Buffer} \xrightarrow{\text{Duration (sec)}} 5$

Connectors with external data sources/sinks

Design overview of TurboProm



TurboProm can provide > 3000x latency reduction



Q1: quantile by (host) (data)

Q2: sum by (host) (data)

Q3:

`quantile_over_time(data[15m])`

Q4: sum by (host)

`(sum_over_time(data[15m]))`



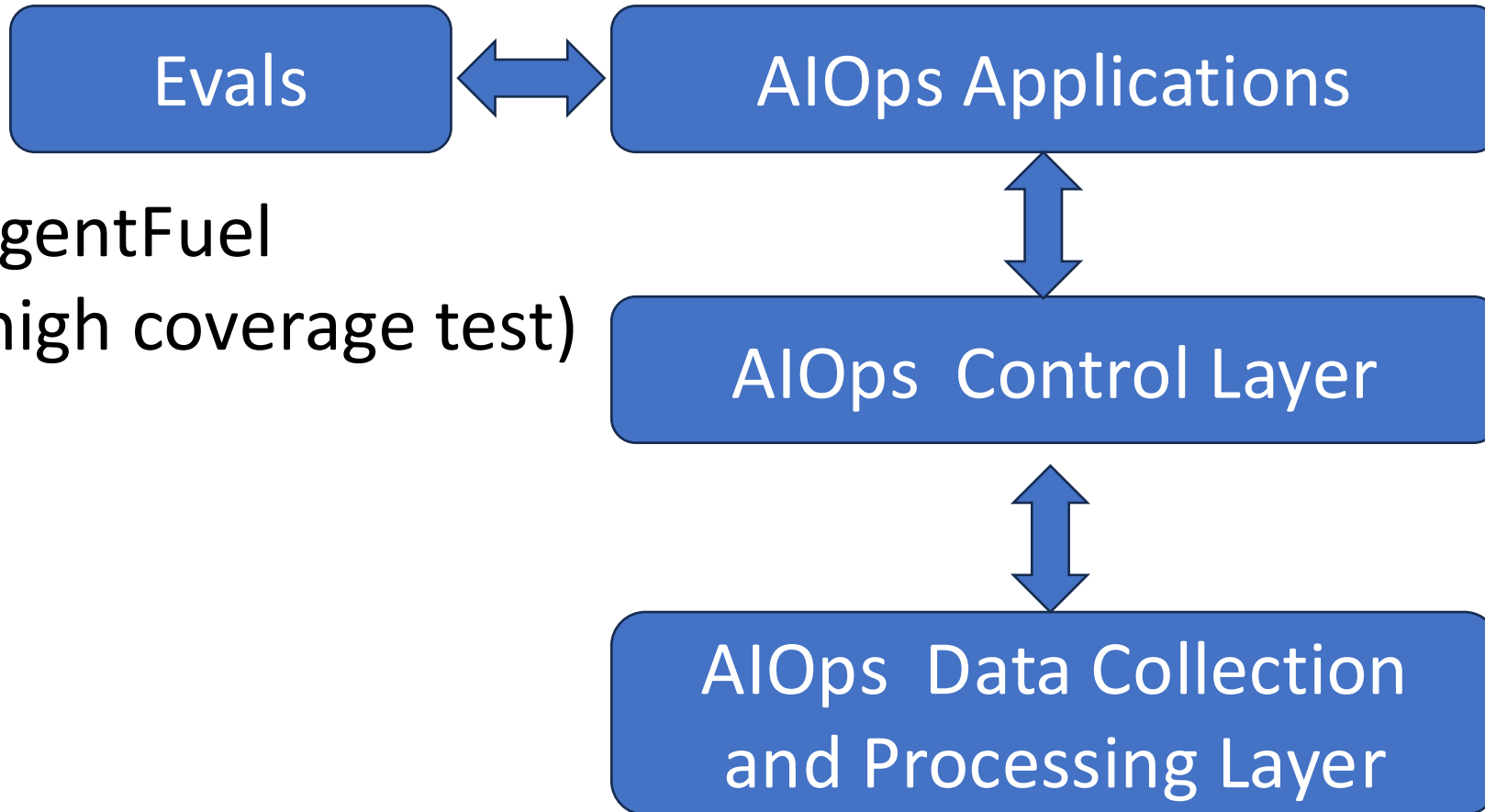
Collect/Transmit/Store: Project ASAP

Unified view of “information summarization” for analytics task

- SketchLib: Fastest sketching library
- SketchPlan: Mapping queries to sketches
- DeepPCAP: Generative models for long term storage of network data
- GenT: Generative models to reduce cost of transmitting traces
- TimeseriesGenModelBench
- New compression primitives

Putting it together: AIOps for the future

Incalmo (red team)
DAGR (Agentic RCA)



MoCE (abstraction)

TimeState (stateful)
TurboProm (approx.)
ASAP (info-summary)

	Key Idea	<i>D1. Avoid rigid solutions; exploit the best of all paradigms</i>	<i>D2. Make explainable solutions a first-class requirement</i>	<i>D3. Make cost-awareness non-negotiable</i>	<i>D4. Explicitly allow for human nudges and intervention</i>	<i>D5. Build better systems not just better algorithms</i>
MoCE	Mixture of Context Aware Experts for RCA	✓	✓	✓	✓	✓
DAGR Agentic RCA	Structured Creativity	✓	✓		✓	
AgentFuel	Systematic eval generation for agents	✓	✓		✓	
TimeState	Custom Stateful Abstraction		✓	✓		
TurboProm	Approximation aware computation			✓		✓
Incalmo	Raise abstraction for red teaming	✓	✓		✓	✓
<i>Your work AIOps and beyond</i>	????	✓	✓	✓	✓	✓

Cyber Autonomy Research Vision

Research philosophy: An *integrative system-wide* approach

AI vendors



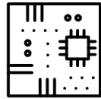
Security vendors



SaaS providers



Hardware vendors



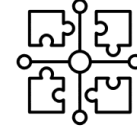
Software developers



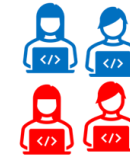
Platform vendors



Systems integrators



Security operators



Governments, NGOs, policymakers

